EXPERIMENT 10

Performance Optimization in Zoho Creator

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Aim

To demonstrate performance optimization techniques in a Zoho Creator application, particularly efficient record retrieval and workflow processing.

Introduction

Performance optimization in low-code applications means designing forms, workflows and scripts so that they use data and processing only as far as needed. Although the platform handles much of the underlying work, workflow design influences how efficiently an application runs. Retrieving more records than required, or repeatedly processing irrelevant records, can reduce efficiency as the number of records and users grows. Efficient workflows retrieve only the data they need and avoid unnecessary processing, supporting scalable and maintainable applications.

Application Used

The practical uses Zoho Creator and an Event Registration application. Deluge scripting is used in a form workflow that notifies a customer when a registration is submitted. The source report notes that the account trial had expired, so upgrade-only features are not claimed as enabled or configured.

Performance Optimization Concept

The key optimization demonstrated is criteria-based record fetching. Instead of retrieving every record from a form, a criterion is used to retrieve only the required record:

```
eventRegistration = Event_Registration[ID == input.ID];
```

The workflow uses input.ID to identify the current record. Only the Event Registration record whose ID matches input.ID is retrieved, avoiding unnecessary retrieval and processing of unrelated records.

Procedure

1. Open the Event Registration application in Zoho Creator in Edit mode.

2. Navigate to Workflow and open Form workflows.

3. Observe the “Notify customers” workflow, triggered on Created → Successful form submission; the source report states it has two actions and is enabled.

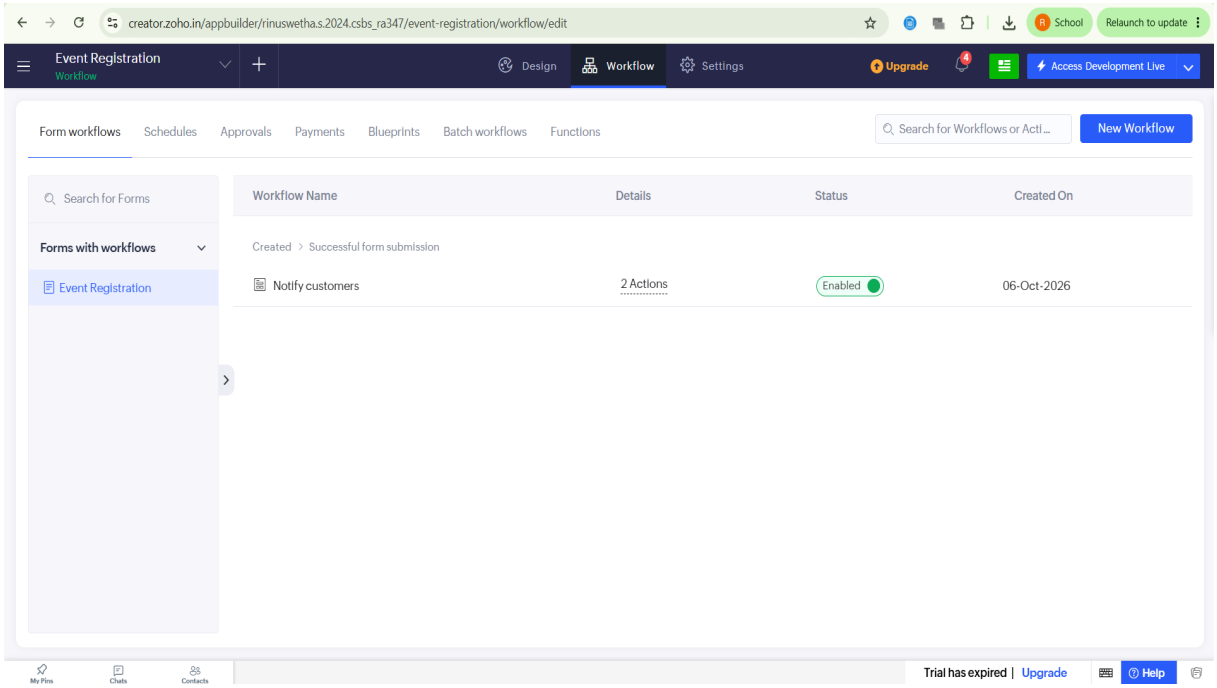
4. When a registration is submitted, identify the current record using input.ID.

5. Fetch the required record using ID == input.ID.

6. Check whether the record’s email field is blank.

7. If an email is available, send a notification; otherwise, display an informational message.

Figure 1: Form Workflow of the Event Registration Application (Actual Screenshot)



Deluge Code

```
eventRegistration = Event_Registration[ID == input.ID];

if(eventRegistration.Email.isBlank())
{
    info "Email address is missing for this registration";
}
else
{
    sendmail
    [
        from :zoho.adminuserid
        to :eventRegistration.Email
        subject :"Your Event Registration is Confirmed"
        message :"<p>Dear Participant,</p><p>Your registration has been received.</p><p>Thank you for registeri
    ]
}
```

The first statement fetches the record by its ID. If the email field is blank, an informational message is displayed. Otherwise, an email is sent to the customer's email address.

Performance Analysis

Approach	Steps
Before (inefficient)	Fetch all records → iterate through records → find the required record → process
Optimized	Fetch the required record using ID → process directly

The inefficient approach retrieves and processes records that are not needed. The optimized approach reduces unnecessary retrieval and processing. No benchmark was performed, so no measured improvement in execution time is claimed.

Optimization Techniques

Technique	Explanation
Criteria-based record fetching	Retrieve only the records required for the current operation. This technique is used in the practical.

Technique	Explanation
Avoid unnecessary iteration	Do not fetch all records and loop through them when a specific criterion can be used.
Efficient built-in functions and aggregate operations	Use built-in functions and aggregate operations instead of unnecessary repeated processing.
Indexed fields	Indexed fields can improve retrieval for frequently queried data. Indexing may require backend/platform configuration and was not enabled in this application.
Efficient workflows	Keep workflows simple and avoid unnecessary nested or repeated operations.

Optimized Workflow

Event Registration Submitted → Identify Current Record → Criteria-Based Record Fetch → Validate Email → Prepare Notification → Send Email

The workflow identifies the current record using input.ID, retrieves only the matching Event Registration record, checks Email.isBlank(), and sends an email if the address is available. Otherwise, an informational message is displayed.

Conclusion

Performance optimization is important in low-code applications because efficiency depends not only on the platform but also on how workflows and scripts are written. In this practical, criteria-based record retrieval using Event_Registration[ID == input.ID] retrieves and processes only the required record. Avoiding unrelated retrieval and unnecessary loops, using built-in operations, and keeping workflows simple can support scalable and maintainable applications. No benchmark was performed, so benefits are described in terms of reduced data retrieval and processing rather than measured execution times.